

Chapter 8

Analysis of Simulation Output

Overview

8.1 Confidence intervals and hypothesis testing

8.2 Estimation Methods

8.3 Simulation Run Statistics

8.4 Replication of Runs

8.5 Elimination of Initial Bias

Nature of Problem

- Once a stochastic variable has been introduced in the simulation model, almost all the system variables describing the system's behavior also become stochastic, so no one measurement can be arbitrarily taken to represent the value of the variable.
- Most of the simulation method presuppose that the observations being made are mutually independent when an experiment is being repeated. However, simulation results are not likely to be independent i.e. a single simulation run will produce many "observed" values of a variable, but the value observed at one time is likely to be influenced by the value at some earlier time.

Nature of Problem

- So the need to account this issue requires careful establishment of statistical method.
- One concern for using statistical method is to ensure that the statistical estimates are consistent i.e. as the sample size increases the estimates tends to true value.
- Another concern is to control bias in measures of both mean values and variances.
- Another concern is to determine how long a simulation should be run in order to obtain a given level of confidence in its results.

Confidence intervals and hypothesis testing

Confidence Interval

- A confidence interval is a range of values that is likely to contain the true value of a performance measure with a specified level of confidence (eg. 95%)
- Simulation results vary because of randomness. A confidence interval indicates how accurate and reliable the estimated result is.
- Instead of reporting a single value, it provides a range within which the true system performance is expected to lie.

Confidence intervals and hypothesis testing

Confidence Interval Formula

$$CI = \bar{x} \pm Z * \frac{s}{\sqrt{n}} \quad \text{For } n \geq 30$$

$$CI = \bar{x} \pm t * \frac{s}{\sqrt{n}} \quad \text{For } n < 30$$

Where:

$\bar{x}$ = Sample mean

Z = Z-score (1.96 for a 95% confidence level)

t = t-value from the t-distribution

s = Sample standard deviation (or σ if known)

n = Number of observations (simulation runs)

Confidence intervals and hypothesis testing

Hypothesis testing

- Hypothesis testing is a statistical method used to determine whether the results obtained from simulation data provide enough evidence to support or reject a claim about a system.
- Simulation results vary because of randomness. Hypothesis testing helps us decide whether these differences are meaningful.
- It helps determine whether the observed differences are real (statistically significant) or simply due to random variation in simulation output.

Confidence intervals and hypothesis testing

Steps of Hypothesis testing

1. State the hypotheses

- **Null Hypothesis (H_0):** Assumes no difference or no effect.
- **Alternative Hypothesis (H_1 or H_a):** Assumes there is a difference or an effect.

2. Choose the Significance Level (α)

- Usually:
 - $\alpha = 0.05$ (5%)
 - $\alpha = 0.01$ (1%)
- It represents the maximum acceptable probability of making a **Type I error**.

Confidence intervals and hypothesis testing

Steps of Hypothesis testing

3. Collect Simulation Data

- Run the simulation multiple times and record the performance measure (e.g., waiting time, throughput, utilization).

4. Calculate the test statistic

- Depending on the problem, use a z-test, t-test, chi-square test, or F-test.

5. Find the p-value or critical value

Compare the calculated test statistic with the critical value, or compare the p-value with α .

6. Make a decision

- If $p\text{-value} \leq \alpha$, reject H_0
- If $p\text{-value} > \alpha$, fail to reject H_0

Estimation Methods

- Estimation is a method used to estimate the true value of a performance measure (such as average waiting time or average queue length) using data collected from a simulation.
- Since we cannot observe an infinite number of simulation results, we use a sample to estimate the true value.
- Usually a random variable is drawn from an infinite population which has a stationary probability distribution with a finite mean μ and finite variance σ^2 .
- This means that the population is not affected by the number of sample variable already made and nor does it change with time.
- If further the value of one sample is not affected in any way by the value of any other sample, the random variables are mutually independent.
- Random variables that meet all these conditions are called independently and identically distributed .
- Central limit theorem must be invoked to rely upon normal distribution of infinite population.
- Only then we can apply estimation method to that variable taken from infinite population.

Estimation Methods

- Random variables that meet all these conditions are said to be independently and identically distributed, usually abbreviated to i.i.d. for which the central limit theorem can be applied.
- The theorem states that the sum of 'n' i.i.d. variables, drawn from a population that has a mean of μ and a variance of σ^2 , is approximately distributed as a normal variable with a mean of $n\mu$ and a variance of $n\sigma^2$.

Estimation Methods

- Any normal distribution can be transformed into a standard normal distribution, that has a mean of 0 and a variance of 1.
- Let x_i ($i=1,2,\dots,n$) be the 'n' i.i.d. random variables. Then normal variate:

$$Z = \frac{\sum_{i=1}^n x_i - n \mu}{\sqrt{n} \sigma}$$

Estimation Methods

- Dividing top and bottom of this fraction by n we get:

- In terms of sample mean $\bar{x}$

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Where,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Simulation Run Statistics

- Consider a single-server system in which the arrivals occur with a Poisson distribution and the service time has an exponential distribution.
- Suppose the study objective is to measure the mean waiting time, defined as the time entities spend waiting to receive service and excluding the service time itself.
- This system is commonly denoted by M/M/1 which indicates; first , that the inter-arrival time is distributed exponentially; second that the service time is distributed exponentially; and, third, that there is one server. The M stands for Markovian, which implies an exponential distribution.

Simulation Run Statistics

- In a simulation run, the simplest approach is to estimate the mean waiting time by accumulating the waiting time of n successive entities and dividing by n .
- This measure, the sample mean, is denoted by $\bar{x}(n)$ to emphasize the fact that its value depends upon the number of observations taken.
- If x_i ($i=1,2,\dots,n$) are the individual waiting times (including the value 0 for those entities that do not have to wait), then

$$\bar{x}(n) = \frac{1}{n} \sum_{i=1}^n x_i$$

Simulation Run Statistics

- Whenever a waiting line forms, the waiting time of each entity on the line clearly depends upon the waiting time of its predecessors.
- Any series of data that has this property of having one value affect other values is said to be autocorrelated.
- The sample mean of autocorrelated data can be shown to approximate a normal distribution as the sample size increases.

Simulation Run Statistics

The equation $\bar{x}(n) = \frac{1}{n} \sum_{i=1}^n x_i$ remains a satisfactory estimate for the mean of autocorrelated data.

A simulation run is started with the system in some initial state, frequently the idle state, in which no service is being given and no entities are waiting.

The early arrivals then have a more than normal probability of obtaining service quickly, so a sample mean that includes the early arrivals will be biased.

Replications of Runs

- Replication of runs is a method in simulation output analysis where the same simulation model is executed multiple times using different random number streams.
- Although the model and input parameters remain the same, different random numbers produce different outputs. Each run is treated as an independent experiment.
- By performing multiple independent runs, we can:
 - Obtain more reliable results.
 - Reduce the effect of randomness.
 - Estimate the true mean and variance.
 - Construct confidence intervals.

Replications of Runs

- The precision of results of a dynamic stochastic system can be increased by repeating the experiment with different random numbers strings.
- For each replication of a small sample size, the sample mean is determined.
- The sample means of the independent runs can be further used to estimate the variance of distribution.
- Suppose that the experiment is repeated “p” times with independent random number series. Let X_{ij} be the i th observation in j th run, then the sample mean and variance for the j th run are:

Replications of Runs

$$\overline{x}_j(n) = \frac{1}{n} \sum_{i=1}^n x_{ij}$$

$$s_j^2 = \frac{1}{n-1} \sum_{i=1}^n [x_{ij} - \overline{x}_j(n)]^2$$

Replications of Runs

- Combining the results of “p” independent measurements gives the following estimates for the mean waiting time, $\bar{x}$, and the variance, s^2 of the population:

$$\bar{x} = \frac{1}{p} \sum_{j=1}^p \bar{x}_j(n)$$

$$s^2 = \frac{1}{p} \sum_{j=1}^p s_j^2(n)$$

Replications of Runs

- The value of $\bar{x}$ is an estimate for the mean waiting time, and s^2 can be used to establish confidence interval, with $p-1$ degree of freedom.
- The length of run of replications is generally selected such that all combined it comes to the sample size N . i.e. $p \cdot n = N$.
- By increasing the number of replications and shortening their length of run, the confidence interval can be narrowed.
- But due to shortening of length of replication the effect of starting conditions will increase.
- The results obtained will not be accurate, especially when the initialization of the runs is not proper. Thus, a compromise has to be made.
- There is no established procedure of dividing the sample size N into replications.
- However, it is suggested that the number of replications should not be very large, and that the sample means should approximate a normal distribution.

Elimination of Initial Bias

Initial Bias

- Initial bias is a condition which occurs when the initial values of the state variables in a simulation do not represent the system's long-run (steady-state) behavior.
- As a result, the simulation output collected at the beginning may be unrealistic and can introduce errors in the estimated performance measures.
- The goal is to select initial conditions that are more representative of the steady-state distribution. This process is called intelligent initialization.

Two ways to specify initial conditions;

1. If the system exists, collect data on it and use these data to specify more typical initial conditions.
 - It requires a large data collection effort.
 - If system being modeled does not exist, this method is impossible to implement.
 - Collecting system data from a second model which is a simplified model is suggested for complex systems.
2. Reduce the impact of initial conditions, by dividing each simulation run into two phases;
 - 1. Initialization phase from time 0 to T_0
 - 2. Data collection phase from time T_0 to stopping time $T_0 + T_E$

Elimination of Initial Bias

- Two general approaches can be taken to remove the bias: the system can be started in a more representative state than the empty state, or the first part of the simulation can be ignored.
- The ideal situation is to know the steady state distribution for the system, and select the initial condition from that distribution.
- In the study previously discussed, repeated the experiments on the M/M/1 system, supplying an initial waiting line for each run, selected at random from the known steady state distribution of waiting line.

Elimination of Initial Bias

- The more common approach to removing the initial bias is to eliminate an initial section of the run.
- The run is started from an idle state and stopped after a certain period of time.
- The run is then restarted with statistics being gathered from the point of restart.
- It is usual to program the simulation so that statistics are gathered from the beginning, and simply wipe out the statistics gathered up to the point of restart.
- No simple rules can be given to decide how long an interval should be eliminated.

Elimination of Initial Bias

- The disadvantage of eliminating the first part of a simulation run is that the estimate of the variance, needed to establish a confidence limit, must be based on less information.
- The reduction in bias, therefore, is obtained at the price of increasing the confidence interval size.

Elimination of Initial Bias

- The ways to eliminate the initial bias are as follows:
 1. Ignore the initial bias occurred during the simulation run.
 2. The system should be started in a representative state than in the empty state.
 3. Start the simulation in the empty state, then stop after initial bias and then start again.
 4. Run the simulation for such a long period of time so that the initial bias has no any significance in the output result.

References

- Geoffrey Gordon, “System Simulation”.